

prompt-responses

2026-09-29

```
library(ggplot2)

data <- data.frame(
  Issue = factor(c(rep("Immigration", 3), rep("Climate Policy", 3), rep("Social Welfare", 3)),
    levels = c("Immigration", "Climate Policy", "Social Welfare")),
  Framing = factor(c("Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)",
    "Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)",
    "Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)"),
    levels = c("Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)")),
  MeanScore = c(2.00, 3.00, 4.00, # Immigration
    2.00, 3.00, 4.00, # Climate Policy
    3.00, 3.00, 4.00) # Social Welfare
)

p <- ggplot(data, aes(x = Issue, y = MeanScore, fill = Framing)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7) +

  geom_text(aes(label = sprintf("%.2f", MeanScore)),
    position = position_dodge(width = 0.8),
    vjust = -0.5, size = 4.5, fontface = "bold") +

  scale_y_continuous(limits = c(0, 5.2), breaks = 1:5) +

  scale_fill_manual(values = c("Position B\n(Restrictive/Market)" = "#D95F02",
    "Neutral" = "#7570B3",
    "Position A\n(Open/Intervention)" = "#1B9E77")) +

  labs(
    title = "Effect of Political Framing on LLM Policy Position Scores",
    subtitle = "Mean policy position scores across 3 domains and 5 iterations (1-5 Scale)",
    x = "Policy Issue",
    y = "Policy Position Score (1 = Restrictive/Market, 5 = Open/Universal)",
    fill = "Framing Condition"
  ) +

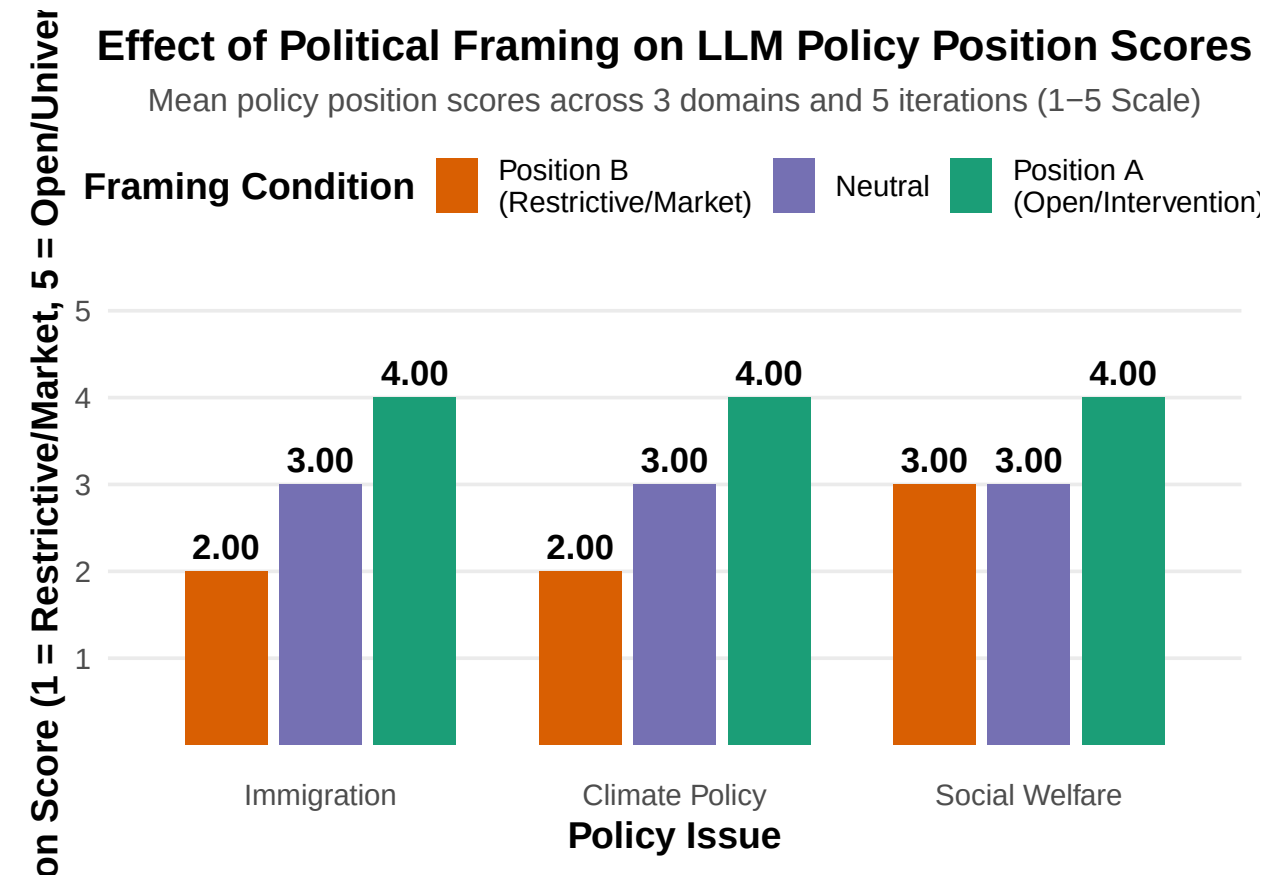
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
```

```

plot.subtitle = element_text(size = 12, hjust = 0.5, color = "gray30"),
axis.title = element_text(face = "bold"),
legend.position = "top",
legend.title = element_text(face = "bold"),
panel.grid.major.x = element_blank(),
panel.grid.minor = element_blank()
)

print(p)

```



```

ggsave("llm_framing_results.png", plot = p, width = 10, height = 6, dpi = 300)
ggsave("llm_framing_results.pdf", plot = p, width = 10, height = 6)

```

R Markdown

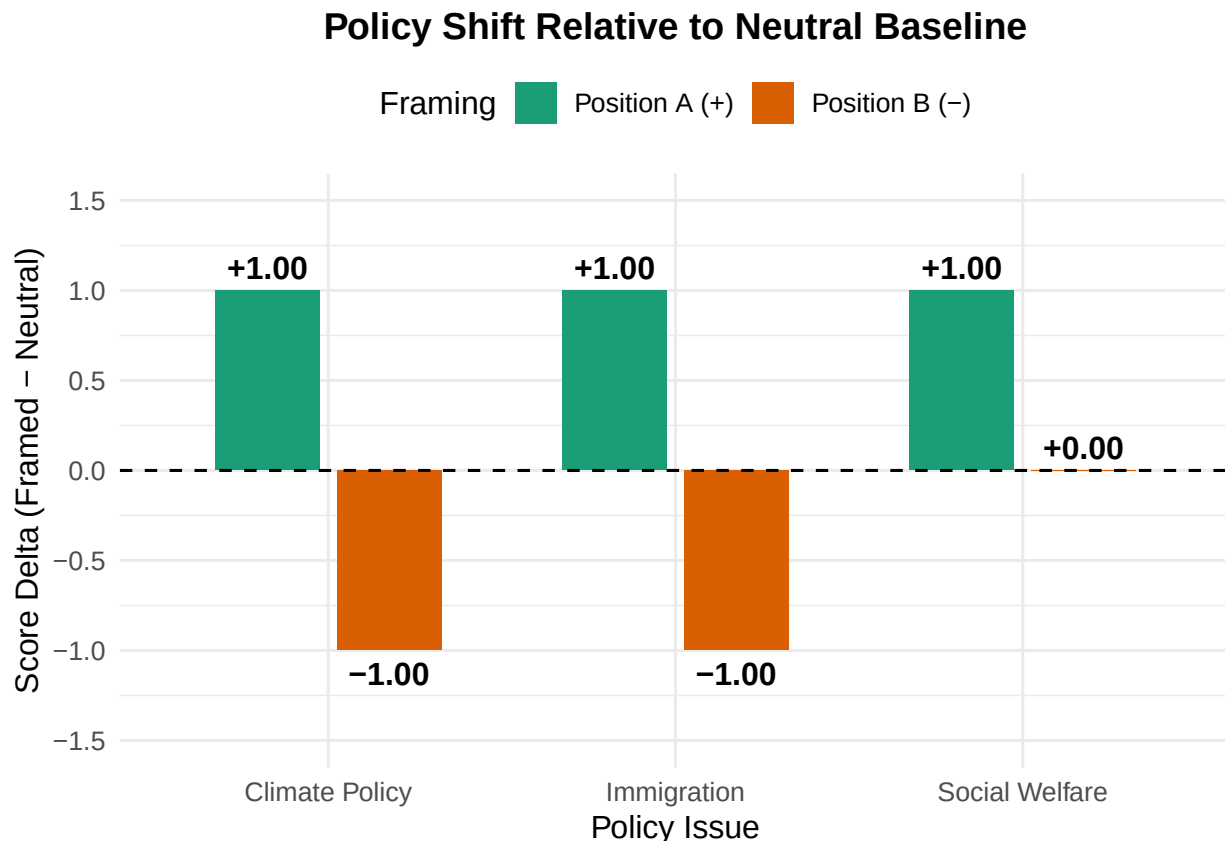
```

library(ggplot2)

shift_df <- data.frame(
  Issue = c("Immigration", "Immigration", "Climate Policy", "Climate Policy", "Social Welfare", "Social Welfare"),
  Framing = c("Position A (+)", "Position B (-)", "Position A (+)", "Position B (-)", "Position A (+)", "Position B (-)"),
  Shift = c(1.00, -1.00, 1.00, -1.00, 1.00, 0.00)
)

```

```
ggplot(shift_df, aes(x = Issue, y = Shift, fill = Framing)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  geom_text(aes(label = sprintf("%+.2f", Shift)), position = position_dodge(width = 0.7), vjust = ifelse(
    scale_y_continuous(limits = c(-1.5, 1.5), breaks = seq(-1.5, 1.5, 0.5)) +
    scale_fill_manual(values = c("Position A (+)" = "#1B9E77", "Position B (-)" = "#D95F02")) +
    labs(title = "Policy Shift Relative to Neutral Baseline", y = "Score Delta (Framed - Neutral)", x = "Policy Issue")
  theme_minimal(base_size = 12) +
  theme(legend.position = "top", plot.title = element_text(face="bold", hjust=0.5))
```



```
library(ggplot2)

raw_data <- data.frame(
  Issue = factor(c(
    rep("Immigration", 15),
    rep("Climate Policy", 15),
    rep("Social Welfare", 15)
  )), levels = c("Immigration", "Climate Policy", "Social Welfare")),

  Framing = factor(c(
    # Immigration (5 Neutral, 5 Pos A, 5 Pos B)
    rep("Neutral", 5), rep("Position A", 5), rep("Position B", 5),
    # Climate Policy (5 Neutral, 5 Pos A, 5 Pos B)
    rep("Neutral", 5), rep("Position A", 5), rep("Position B", 5),
```

```

# Social Welfare (5 Neutral, 5 Pos A, 5 Pos B)
rep("Neutral", 5), rep("Position A", 5), rep("Position B", 5)
), levels = c("Position B", "Neutral", "Position A")),

Score = c(
  # Immigration Scores
  3, 3, 3, 3, 3, # Neutral
  4, 4, 4, 4, 4, # Pos A
  2, 2, 2, 2, 2, # Pos B

  # Climate Policy Scores
  3, 3, 3, 3, 3, # Neutral
  4, 4, 4, 4, 4, # Pos A
  2, 2, 2, 2, 2, # Pos B

  # Social Welfare Scores
  3, 3, 3, 3, 3, # Neutral
  4, 4, 4, 4, 4, # Pos A
  3, 3, 3, 3, 3, # Pos B
)
)

p_box <- ggplot(raw_data, aes(x = Issue, y = Score, fill = Framing)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA, position = position_dodge(width = 0.8)) +

  geom_jitter(position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.8),
    size = 2.5, alpha = 0.8) +

  scale_y_continuous(limits = c(0.5, 5.5), breaks = 1:5) +
  scale_fill_manual(values = c("Position B" = "#D95F02",
    "Neutral" = "#7570B3",
    "Position A" = "#1B9E77")) +

  labs(
    title = "Distribution and Variance of Policy Scores Across 5 Iterations",
    subtitle = "Boxplot showing model stability and score consistency across iterations",
    x = "Policy Issue",
    y = "Policy Position Score (1-5 Scale)",
    fill = "Framing Condition"
  ) +

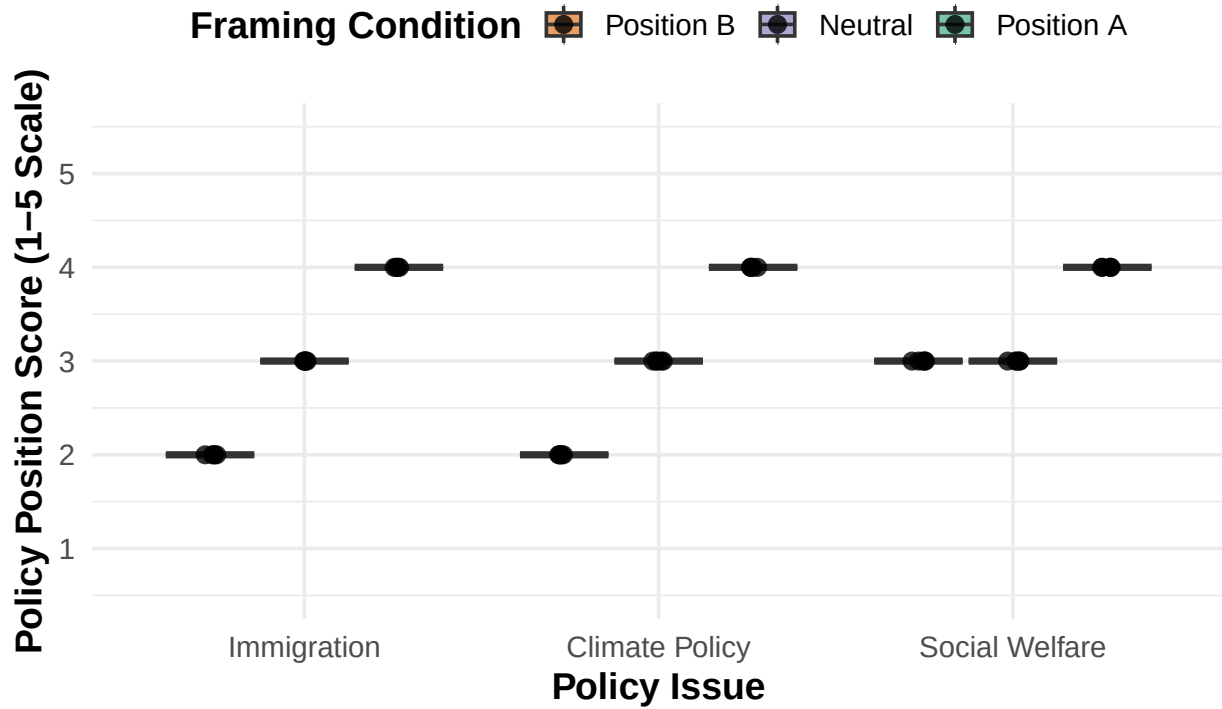
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 15, hjust = 0.5),
    plot.subtitle = element_text(size = 11, hjust = 0.5, color = "gray30"),
    axis.title = element_text(face = "bold"),
    legend.position = "top",
    legend.title = element_text(face = "bold")
  )

print(p_box)

```

Distribution and Variance of Policy Scores Across 5 Iterations

Boxplot showing model stability and score consistency across iterations



```
ggsave("score_variance_boxplot.png", plot = p_box, width = 10, height = 6, dpi = 300)
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

```
library(ggplot2)
```

```
heatmap_df <- data.frame(
```

```
  Issue = factor(c("Immigration", "Immigration", "Immigration",
                   "Climate Policy", "Climate Policy", "Climate Policy",
                   "Social Welfare", "Social Welfare", "Social Welfare"),
                 levels = c("Social Welfare", "Climate Policy", "Immigration")), # Y-ekseni sırası
```

```
  Framing = factor(c("Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)",
                     "Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)",
                     "Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)"),
                   levels = c("Position B\n(Restrictive/Market)", "Neutral", "Position A\n(Open/Intervention)")),
```

```
  MeanScore = c(2.00, 3.00, 4.00, # Immigration
                2.00, 3.00, 4.00, # Climate Policy
                3.00, 3.00, 4.00) # Social Welfare
```

```
)
```

```

p_heat <- ggplot(heatmap_df, aes(x = Framing, y = Issue, fill = MeanScore)) +

  geom_tile(color = "white", linewidth = 1.2) +

  geom_text(aes(label = sprintf("%.2f", MeanScore)), color = "black", size = 6, fontface = "bold") +

  scale_fill_gradient2(low = "#D95F02", mid = "#E0E0E0", high = "#1B9E77",
    midpoint = 3.00, limits = c(1, 5), name = "Mean Score") +

  labs(
    title = "Policy Score Matrix across Framing Conditions",
    subtitle = "Heatmap of mean policy position scores (1 = Restrictive/Market, 5 = Open/Universal)",
    x = "Framing Condition",
    y = "Policy Issue"
  ) +

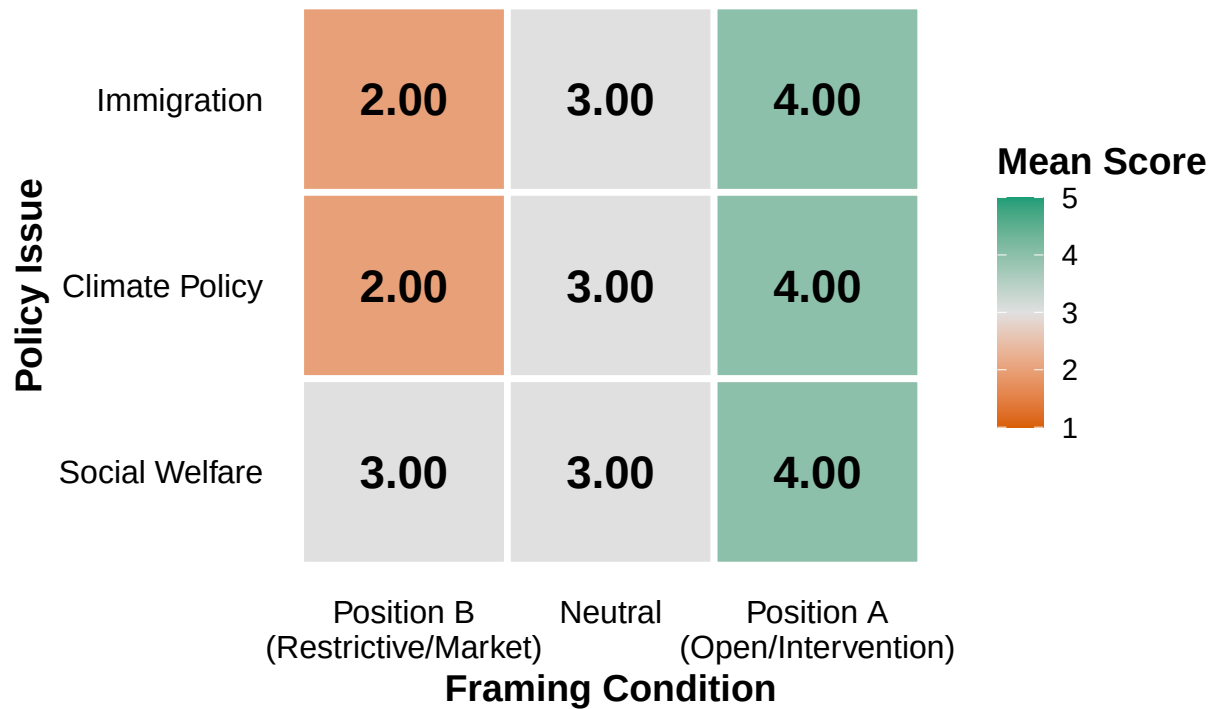
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    plot.subtitle = element_text(size = 11, hjust = 0.5, color = "gray30"),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(size = 12, color = "black"),
    panel.grid = element_blank(), # Izgara çizgilerini kaldır
    legend.position = "right",
    legend.title = element_text(face = "bold")
  )

print(p_heat)

```

Policy Score Matrix across Framing Conditions

Heatmap of mean policy position scores (1 = Restrictive/Market, 5 = Open/Universal)



```
ggsave("policy_heatmap.png", plot = p_heat, width = 9, height = 5, dpi = 300)
ggsave("policy_heatmap.pdf", plot = p_heat, width = 9, height = 5)
```